Data Communication

Content 05: Analog Transmission

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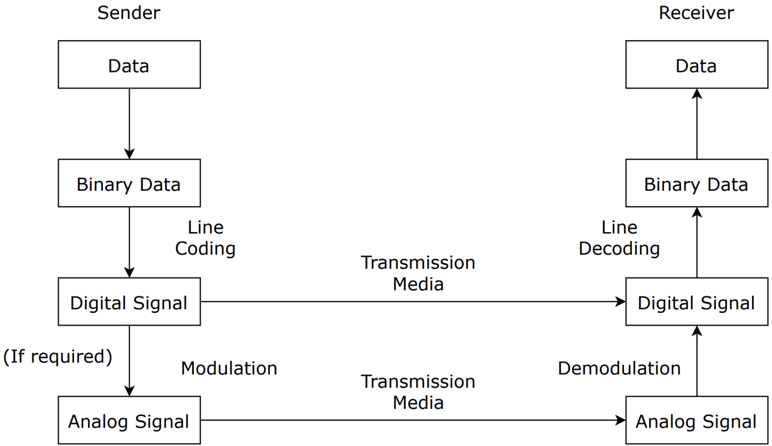
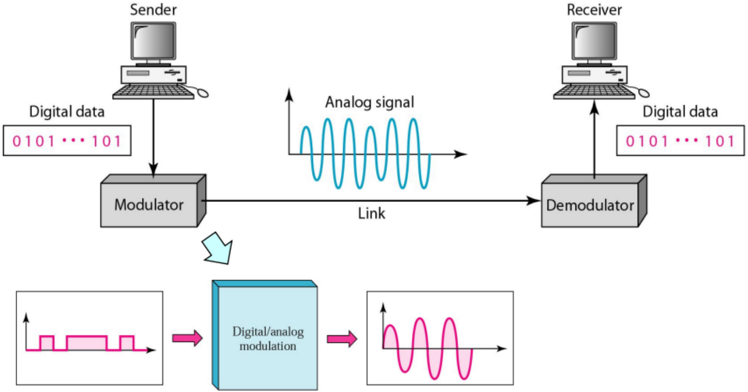
Data Transmission Pipeline

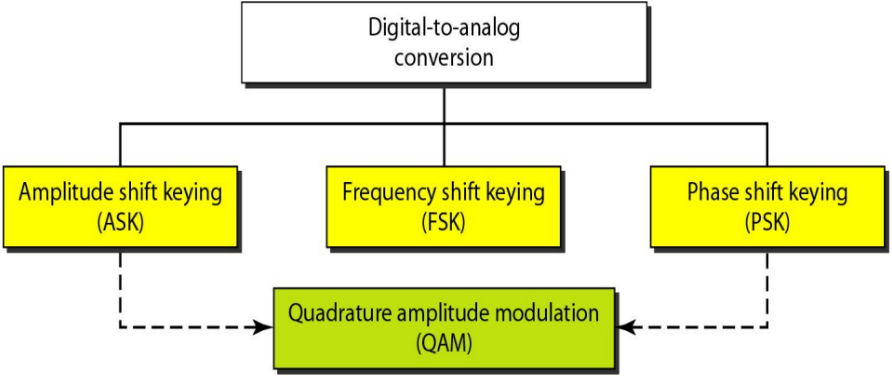


Digital-To-Analog Conversion

Digital-to-analog conversion is the process of changing one of the characteristics of an analog signal based on the information in digital data.

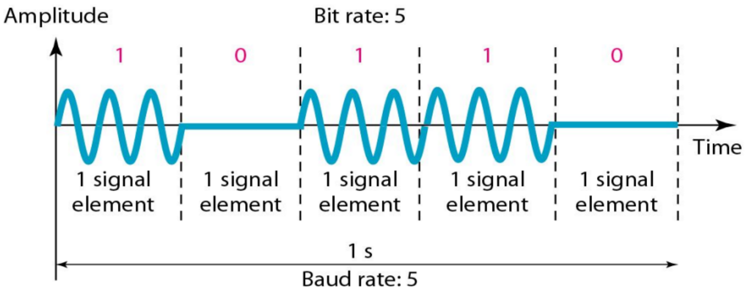


Classification

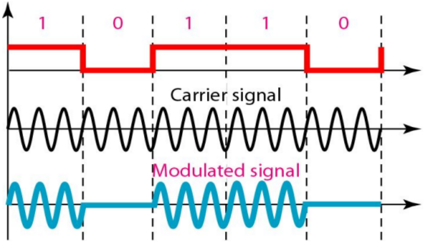


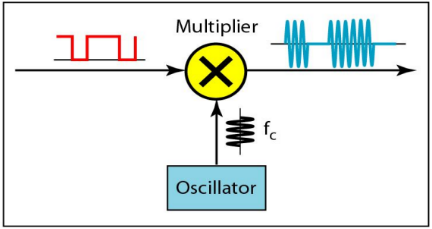
Amplitude Shift Keying (ASK)

- The amplitude of the carrier signal is varied to create signal element. Both frequency and phase remain constant while the amplitude changes.
- Highly susceptible to noise interference.



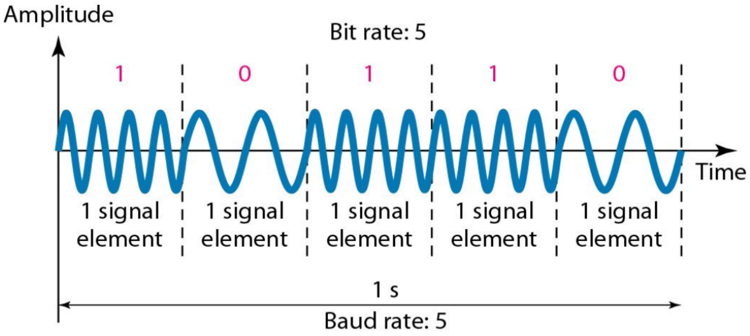
Amplitude Shift Keying (ASK) Implementation



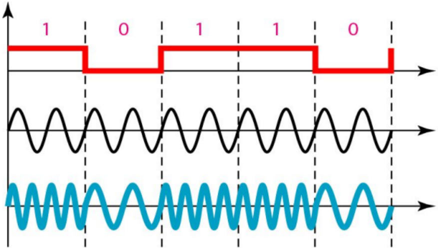


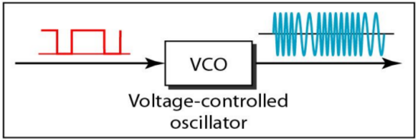
Frequency Shift Keying (FSK)

- The frequency of the carrier signal is varied to represent binary 1 or 0.
- Peak amplitude and phase remain constant.



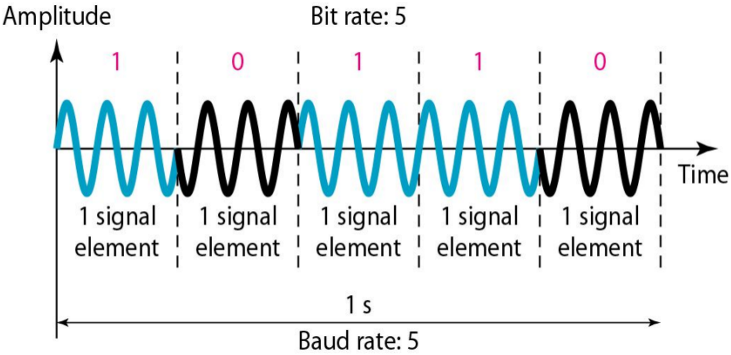
Frequency Shift Keying (FSK) Implementation



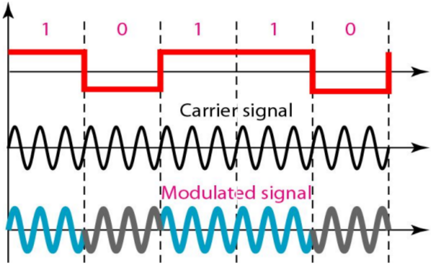


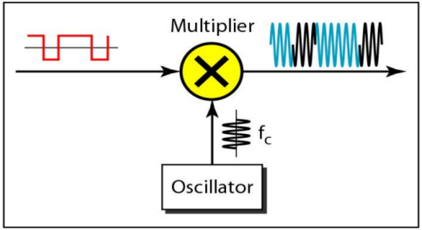
Phase Shift Keying (PSK)

- The phase of the carrier is varied to represent two or more different signal elements.
- Both peak amplitude and frequency remain constant as the phase changes.



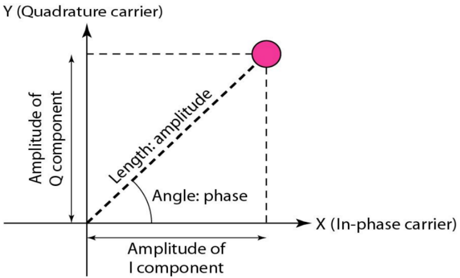
Phase Shift Keying (PSK) Implementation





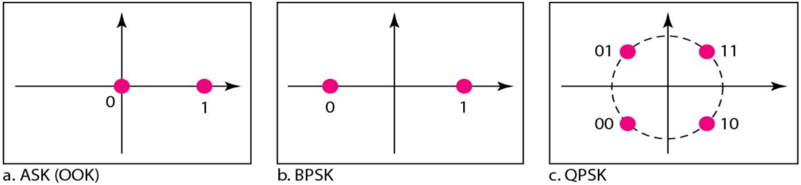
Constellation Diagram

- **A constellation diagram** is a representation of a modulated signal in the complex plane, where each point (called a symbol) represents a unique combination of amplitude and phase used in modulation schemes.
- **The distance between points** on the diagram determines the system's noise immunity — greater distance between symbols means better error performance, as it reduces the chance of misinterpretation due to noise.



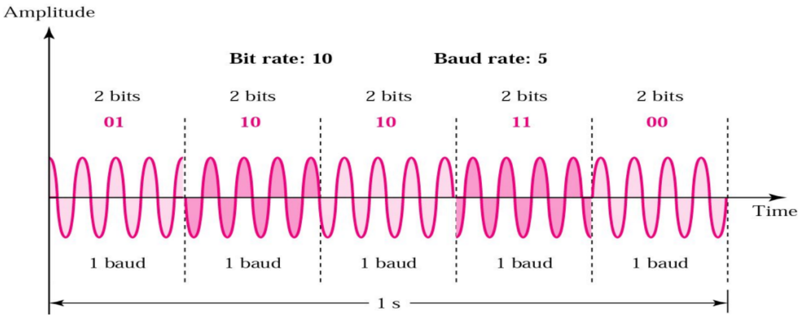
Constellation Diagram

- **ASK:** Points appear along a line, differing only in amplitude.
- **BPSK:** Two points on a circle, 180° apart, showing phase shift.
- **QPSK:** Four points spaced at 90° angles, each encoding 2 bits.



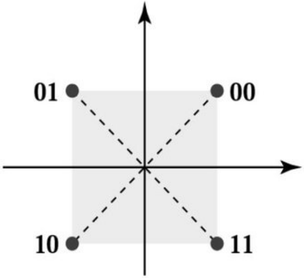
Quadratic Phase Shift Keying (QPSK)

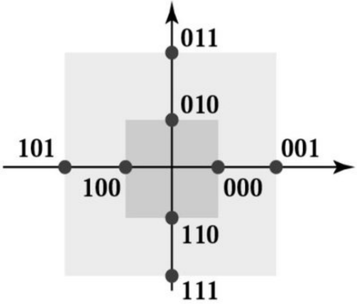
- Instead of utilizing only two variations of a signal, we can use 4 variations and let each phase shift represent 2 bits.



Quadrature Amplitude Modulation (QAM)

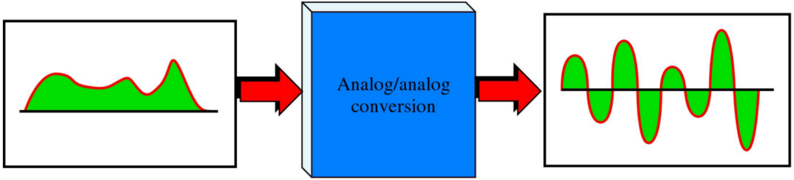
- **Why use QAM:** It increases data rate by sending multiple bits per symbol.
- **How it works:** Combines amplitude and phase variations to form unique symbols.
- **Constellation:** Points are arranged in a grid pattern on the I-Q plane.



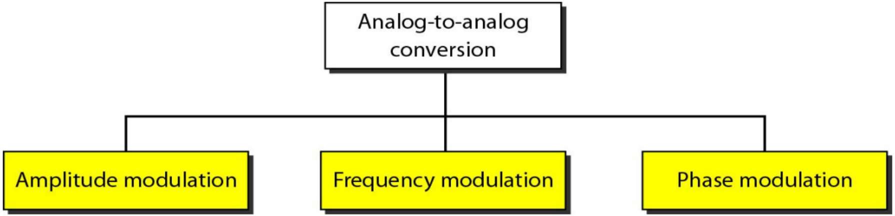


Analog-To-Analog Conversion

Analog-to-analog conversion modifies an analog signal's characteristics (amplitude, frequency, or phase) to transmit information.

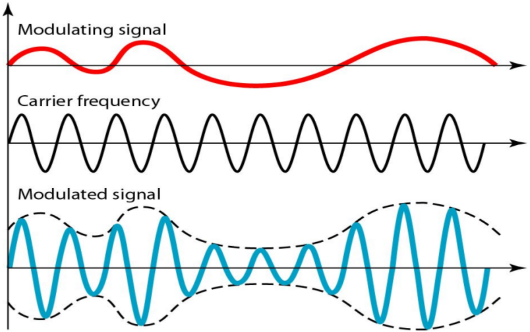


Analog-To-Analog Conversion



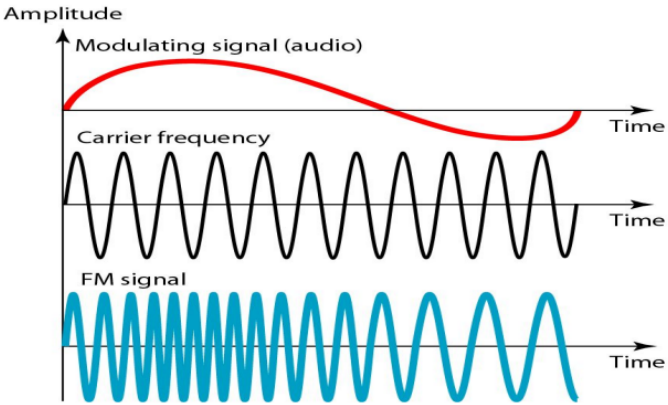
Amplitude Modulation (AM)

- The frequency and phase of the carrier remain the same.
- The amplitude changes to follow variations in the information.



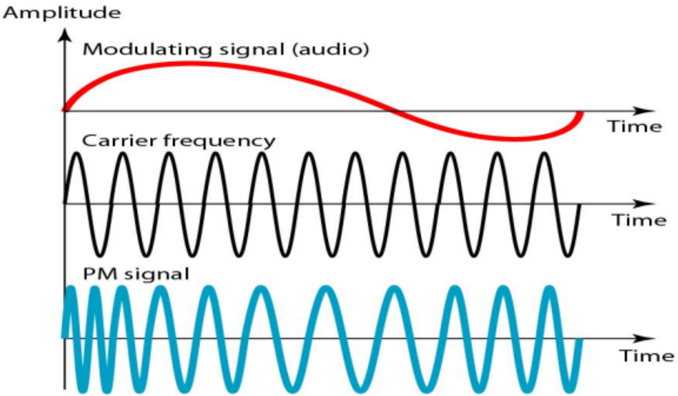
Frequency Modulation (FM)

- As the amplitude of the information signal changes, the frequency of the carrier changes proportionately.



Phase Modulation (PM)

- The phase of the carrier signal is modulated to follow the changing voltage (amplitude) of the modulating signal.



Practice Problems

- Theoretical:
  Data Communication and Networking (4th Ed) - Behrouz A. Forouzan
  Chapter 05 - 3, 4, 6, 8, 9

References

- Data Communication and Networking (4th Ed) - Behrouz A. Forouzan
  Chapter 05 - 5.1, 5.2

Thank You